

Quantum Gravitation

Lecture 2

Normal Modes of the Gravitational Medium

- Degrees of freedom
- Quantization
- Spectrum and response
- Consistency checks
- Example

■ Degrees of freedom

We separate propagating polarizations from constrained components before quantization.

$$h_{ij}(x) = \sum_{k,s} q_{ks} e_{ij}^s(k) e^{ikx}$$

■ Quantization

Canonical normalization is fixed before creation and annihilation operators are introduced.

$$H = \frac{1}{2} \sum_{k,s} (p_{ks}^2 + \omega_{ks}^2 q_{ks}^2)$$

■ Spectrum and response

Poles, residues, and group velocity are read from the same quadratic operator.

$$\mathcal{E}_{ij,kl} e_{kl}^s = \rho_m \omega_s^2 e_{ij}^s$$

■ Consistency checks

Hermiticity, positive norm, source conservation, and the low-energy limit are checked explicitly.

■ Example

A short numerical estimate fixes the scale of the effect before a full fit is attempted.

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Reading: Seo, ch. 2; Mode Atlas MA-47; Cole, ch. 3